Algorithm for Bioinformatics Exercise 5 David Riemer, Usama Mehmood

Problem 1

In this exercise we have the following parameters:

2 Strings: $A[1..m], B[1..n]$, the matching/mismatching costs $p(a, b) \geq 0$ and the opening costs g_0, as well as the gap extension costs g with the goal of minimizing the overall total cost. We can set the opening costs to some number, e.g. 3 and the extension costs to 1. Other to the algorithm in class we use 3 different matrices to store the information, if we are opening a new gap, or extending an already existing one. We use the 3 matrices M, X, Y.

So at each position (i, j)

> The matrix M symbolizes diagonal downwards movement and a match/mismatch state and consumes both $A[i]$ and $B[j]$. So every gap that was opened is closed as well.
>
> $X(i, j)$ consumes $A[i]$ only and represents a gap in B. X can extend an already opened block, or open a new gap block from M or Y.
>
> Alternatively Y represents a horizontal movement and consumes $B[j]$ only. Y can also extend an already opened bloc with $Y[i, j-1] + g$ or open a new gap from M or X with $M \; or \; X + g_0 + g$.
>
> If we are already inside of a gap, we only add g to the distance of the strings and if we open a new are starting a new gap block we are adding $g_0 + g$.

So:

$$M(i,j) = \min\big(M(i-1,j-1),\; X(i-1,j-1),\; Y(i-1,j-1)\big) + p(A[i], B[j])$$

$$X(i,j) = \min\big(M(i-1,j) + g_0 + g,\; X(i-1,j) + g,\; Y(i-1,j) + g_0 + g\big)$$

$$Y(i,j) = \min\big(M(i,j-1) + g_0 + g,\; X(i,j-1) + g_0 + g,\; Y(i,j-1) + g\big)$$

For the global alignment the cost and final answer then is

$$cost = min\{M[m,n], X[m,n], Y[m,n]\}$$

Initialization

The first rows and columns get initialized as follows:

$M[0,0] = 0, X[0,0] = Y[0,0] = +\infty$

$X[i,0] = g_0 + i * g,\ (i \geq 1),\ M[i,0] = +\infty, Y[i,0] = +\infty$

$Y[0,j] = g_0 + j * g,\ (j \geq 1),\ M[0,j] = +\infty, X[0,j] = +\infty$

Example:

We compare the 2 sequences ABC and A.

The 3 matrices have the form of 4×2. After initializing the matrices we get

$$M = \begin{pmatrix} 0 & \infty \\ \infty & \\ \infty & \\ \infty & \end{pmatrix} X = \begin{pmatrix} \infty & \infty \\ 4 & \\ 5 & \\ 6 & \end{pmatrix} Y = \begin{pmatrix} \infty & 4 \\ \infty & \\ \infty & \\ \infty & \end{pmatrix}$$

The value 4 means opening a new gap (length 1), the value 5 means extending that same gap to length 2 and 6 means extending it to length 3.

Filling in the interior:

This is computed with the rules above. So for example the cell $(1,1)$ is calculated with

$M[1,1] = min(M(0,0), X(0,0), Y(0,0)) + p(A[i], B[j])$
$= min(0, +\infty, +\infty) + 0 = 0$

After filling in the cells with the rules above, we come out to the matrices:

$$M = \begin{pmatrix} 0 & \infty \\ \infty & 0 \\ \infty & 6 \\ \infty & 7 \end{pmatrix} X = \begin{pmatrix} \infty & \infty \\ 4 & 8 \\ 5 & 4 \\ 6 & 5 \end{pmatrix} Y = \begin{pmatrix} \infty & 4 \\ \infty & 8 \\ \infty & 9 \\ \infty & 10 \end{pmatrix}$$

So the solution would be 5.

Each cell is omputed in $O(1)$ time and we have matrices of size $(m+1) \times (n+1)$ so the total runtime is also $O(mn)$.

Problem 2

a)

We have the following 4 axioms $M1 - M2$ that describe a metric. We need to show that the sim_s fullfills these axioms over $\sum^*$.

M1

From the slides we defined that the alignments are either of the types $(a, b), (a, -)\ or\ (-, b)$.

With the costs of $p(a, b)$ or g.

- symmetric matches/substitutions matrix $p : \Sigma \times \Sigma \to \mathbb{R}$ $\quad (p(a,b) = p(b,a))$
- gap penalty $g \in \mathbb{R}$

$\rightsquigarrow$ $S\left(\begin{bmatrix} c \\ c' \end{bmatrix}\right) = p(a,b)$, $S\left(\begin{bmatrix} c \\ - \end{bmatrix}\right) = S\left(\begin{bmatrix} - \\ c \end{bmatrix}\right) = g$

$\rightsquigarrow$ score of alignment sum of scores of columns

Since p is also a metric we can assume that $p(a, b) \geq 0$. So is g, given from the task. So every alignment has to be ≥ 0, so tehir sum as well.

M2

If the 2 strings are the same, the alignment score is also 0. Indeed $sim_s = 0 \iff x = y$

Since $sim_s = 0$ and $g > 0$, no gaps can appear in the string. Also $p(a, b)$ must also always be 0, so every character must be the same. So indeed $a = b$ for every column. This means, that the 2 strings have the same lenghts as well as the samee characters as well.

The other direction $x = y \Rightarrow sim_s = 0$ is trivial.

M3

We need to prove that $sim_s(x, y) = sim_s(y, x)$. This is true, since (also from the slides) $p(a, b) = p(b, a)$, as well as a column $(a, -)$ becomes $(-, a)$. These also have the same gap cost g. So for any alignment the reverse cost of flipping x and y also exists, as well as their minima.

M4

We need to show that

$$sim_S(x,z) \leq sim_S(x,y) + sim_S(y,z), \forall x,y,z \in \sum^*$$

b)

We choose the 3 letter words a, b, c and define:

$$p(a,a) = 0, p(b,b) = 0, p(c,c) = 0, p(a,b) = 1, p(b,c) = 1, p(a,c) = 3,$$

This definition violates M4 with $3 > 2$.

The gap penalty g in our example is also $= 1$

So now, if we want sim_s for all the possibilities, we find that it conserves all the axioms given by the task.

$$\begin{gathered} sim_s(a,b) = p(a,b) = 1 \\ sim_s(b,c) = p(b,c) = 1 \\ sim_s(a,c) = 2 * g = 2 \end{gathered}$$

$\Rightarrow$Conserves M4.

c)

The difference to a) is that now we also handle affine gap costs.

From our slides this is as defined as follows:

⇝ *affine gap costs:*
score k contiguous insertions (or k contiguous deletions) instead as $g_0 + k \cdot g$
(usually then $g_0 \gg g$)

- If we represent contiguous insertions as $\begin{bmatrix} \vdash \\ c_1 \end{bmatrix} \begin{bmatrix} - \\ c_2 \end{bmatrix} \cdots \begin{bmatrix} - \\ c_k \end{bmatrix}$
can assign $S(\begin{bmatrix} \vdash \\ c \end{bmatrix}) = g_0 + g$ and $S\begin{bmatrix} - \\ c \end{bmatrix} = g$.

M1

Like in a), since p is also a metric we can assume that $p(a, b) \geq 0$. Since $g_0, g \geq 0$ and also $g_0 * g > 0$, so sim_S also has to be ≥ 0.

M2

If we had a gab block, g_0 would be greater than 0. If we had any mismatch qith $a \neq b$ the cost $p(a, b)$ would be > 0.

Since $sim_s = 0$ and $g > 0$, no gaps can appear in the string. Also $p(a, b)$ must also always be 0, so every character must be the same. So indeed $a = b$ for every column. This means, that the 2 strings have the same lenghts as well as the samee characters as well.

The other direction $x = y \Rightarrow sim_s = 0$ is trivial.

M3

Is also still true and unaffected by the the insertion or deletion of a block of length k, so the cost of $g_0 + kg$ is unaffected.

M4

d)

We look at the following added script operations.

$\rightsquigarrow$ Allow new edit script operations (all cost 0):

- IgnorePrefix($A[0..i]$) free deletes at beginning
- IgnorePrefix($B[0..j]$) free inserts at beginning
- IgnoreSuffix($A[i..m]$) free deletes at end
- IgnoreSuffix($B[j..n]$) free inserts at end

Even if the 2 strings are not the same, their local distance can be 0, breaking M2.

Example

$$x = A$$
$$y = XAX$$
$$d_{local}(x, y) = 0$$